

Residual $SO(4, 2)$ Cohomology of Free Weyl Gravity on the Conformal Cylinder: Cartan Homotopy, Weyl-Square Classes, and Their Residual Pairing

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Abstract

Treating all fifteen cylinder conformal transformations, including compact time translation D , as residual gauge transformations, we compute a minimal absolute $SO(4, 2)$ Chevalley–Eilenberg complex for free four-dimensional Weyl gravity on $\mathbb{R} \times S^3$. Its coefficient module is the parity-complete on-shell Weyl-curvature Fock module. The standard invariant one-particle form is indefinite, with cylinder signs $+I_E \oplus (-I_A) \oplus (-I_L)$.

The key mechanism is the Cartan homotopy formula. The residual differential obeys $d\iota_D + \iota_D d = \mathcal{L}_D$, so every subcomplex of nonzero total compact degree is contractible. In the Hamada-centered four-ghost polarization, the selected absolute degree has ghost compact degree bounded below by -4 , and only matter weights at most four can contribute. The vacuum and one-particle sectors are acyclic, while a cutoff-complete absolute calculation in the lowest two-particle sector gives

$$H_{\text{res}, \text{min}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{rep}} = I_2.$$

Here $G_{\text{rep}} = I_2$ is the normalized Hermitian Gram matrix of the two residual representatives. The survivors are ghost-dressed vertex or deformation classes, not one-particle graviton states. Descent identifies their parity-even and parity-odd combinations with $C_{\mu\nu\rho\sigma} \tilde{C}^{\mu\nu\rho\sigma}$ and $C_{\mu\nu\rho\sigma} \tilde{C}^{\mu\nu\rho\sigma}$. The first varies to the Bach equation; the second is the Pontryagin boundary direction. After quotienting variationally trivial bulk functionals, the local dynamical deformation space is one-dimensional and has representative Gram matrix I_1 .

This algebraic residual theorem is exact once the on-shell Weyl module, the Hamada ghost polarization, and full residual conformal gauging are assumed. The last assumption is a physical choice: if D is retained as a Hamiltonian or boundary charge, the contraction is not a physical quotient. We separately formulate the hypotheses under which the result transfers to the complete free strict-pure-Weyl BV complex. Formal detour identities and exact homogeneous-jet calculations through degrees two to six support that bridge, but the all-level Lorentzian cylinder-kernel theorem and the full zero-mode transfer remain open. We claim neither a positive graviton Fock space nor quantum anomaly freedom.

Contents

1 Introduction

2

2	The free local pure-Weyl complex	4
2.1	Gauge, curvature, and equation maps	4
2.2	What the local identities do not yet prove	5
2.3	A prospective Lorentzian proof route	6
3	The on-shell Weyl module	7
3.1	Character and cylinder towers	7
3.2	Invariant form and exact generator action	7
4	Hamiltonian moment-map normalization	8
5	Residual BRST and Cartan homotopy reduction	9
5.1	The absolute residual complex	9
5.2	Full residual gauging is a physical assumption	9
5.3	Cartan homotopy	10
5.4	Finite centered inventory	10
6	Complete centered cohomology	11
6.1	The relative weight-four kernel	11
6.2	Absolute kernel and image	12
6.3	The normalized residual representative Gram matrix	12
7	Descent and the dynamical/topological split	13
7.1	Why weight four is selected	13
7.2	Parity basis	14
7.3	Euler–Lagrange quotient	14
8	The local-to-residual transfer	15
8.1	Equivariant Cartan transfer	15
8.2	Cyclic transfer and the pairing	15
8.3	The filtered local-to-residual complex	16
8.4	Pure-Weyl field-theoretic hypotheses	17
9	Scope, interpretation, and open questions	17
10	Conclusion	18
A	Conventions	18
B	Exact rank certification	19
C	Claim-status and verification index	19

1 Introduction

Pure conformal gravity in four dimensions is defined, up to an overall coupling and the Euler density, by the Weyl-square action

$$S_W[g] = -\alpha_g \int_M d^4x \sqrt{-g} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}. \quad (1)$$

Its fourth-order linearized equation has more solutions than the Einstein equation, and a conventional local reduction produces an indefinite oscillator form. This familiar observation does not by itself determine the state space after all gauge constraints have been imposed. On a compact spatial slice, background conformal symmetries are reducibilities of the local $\text{Diff} \times \text{Weyl}$ gauge system. Treating those residual transformations as gauge introduces a second, finite-dimensional reduction which is absent from a naive oscillator count. It is also a substantive physical choice. In a Dirac or background-independence interpretation of a closed universe, all residual conformal transformations can be quotiented. In a fixed cylinder conformal field theory, by contrast, D is ordinarily the Hamiltonian and the conformal group acts globally. This paper computes the first choice; it does not assert that strict pure-Weyl gravity uniquely selects it.

The organizing question is therefore more basic than a sign assignment: before asking whether a gravitational mode has positive or negative norm, one must first determine whether it survives the full gauge reduction being imposed. The exact answer below concerns the residual reduction; the bridge from that calculation to the complete local BV theory remains conditional.

The present paper isolates that free classical problem on

$$M = \mathbb{R} \times S^3, \quad \bar{g} = -dt^2 + d\Omega_3^2, \quad (2)$$

with unit cylinder radius. Three spaces must be distinguished:

$$\begin{aligned} \text{local oscillator cohomology} &\supseteq \text{classically integrable perturbations}, \\ \text{classically integrable perturbations} &\longrightarrow \text{residual-BRST classes}. \end{aligned} \quad (3)$$

The first contains the familiar Einstein, vector-descendant, and logarithmic or higher-derivative towers. The last is obtained only after the residual $SO(4, 2)$ constraints and their ghosts are included. A linearized oscillator need not survive this final reduction.

Our result is most naturally stated in the language of residual vertex cohomology. The Hamada ghost polarization supplies a top-degree four-ghost factor of compact degree -4 . It pairs with scalar matter primaries of weight four. The complete centered residual cohomology consists of two chiral Weyl-square classes, and their normalized residual representative Gram matrix is positive. Their parity combinations are the dynamical Weyl-square density and the topological Pontryagin density. Thus the positive matrix is attached to classical vertex or theory-space directions. It is not a positive norm on a propagating graviton Fock space; indeed, the one-particle residual cohomology vanishes.

Relation to previous work. The conformal deformation detour complex on obstruction-flat backgrounds was constructed by Branson and Gover [1]; its formal self-adjointness and related Yang–Mills detour constructions were developed in [2]. Cylinder harmonics, oscillator charges, and primary building blocks were developed by Hamada and Horata and by Hamada [5, 6]. In the broader conformal-mode/Riegert programme, Hamada identified the residual diffeomorphisms with cylinder conformal transformations [7], constructed the fifteen-generator residual BRST charge, used the product of four proper-conformal ghosts as the centered ghost vacuum, derived the scalar weight-four condition, and displayed a Weyl-square BRST state [8]. None of those ingredients is claimed as new here. We use the harmonic and residual-algebra conventions, but we do not import the Riegert field or the quantum claims of that model. The Weyl-curvature module and its character are consistent with the general conformal higher-spin description and partition-function literature [9, 10]. Finally, the identification of C^2 as the unique local four-derivative self-interaction of one linear conformal graviton, under the assumptions stated there, is due to Boulanger and Henneaux [11].

The defensible new result is narrower and more concrete. Starting with the parity-complete pure-Weyl coefficient module, we compute the complete minimal *absolute* residual complex in the centered degree. We prove that the vacuum and one-particle sectors vanish, that precisely two two-particle classes survive after the incoming-image test, and that their normalized Hermitian representative Gram matrix is I_2 . The standard Cartan homotopy makes the particle-number inventory exhaustive, and the Euler–Lagrange map splits the two positive directions into one dynamical and one topological class.

Claim boundary. Three statements must remain separate.

1. The *algebraic residual theorem* is exact for the minimal absolute $SO(4, 2)$ Chevalley–Eilenberg complex built on the on-shell Weyl module, the stated centered ghost polarization, and full residual conformal gauging.
2. The identification of that algebraic answer with the complete free strict-pure-Weyl local-plus-residual BV cohomology is conditional. It requires an all-level Lorentzian local-kernel theorem, a conformal-Killing zero-mode split, and a compact-degree-equivariant cyclic transfer. The formal identities and finite exact tests are given here; their remaining field-theoretic hypotheses are displayed explicitly rather than hidden.
3. Calling the resulting classes physical states additionally requires a boundary and gauge principle which makes all fifteen conformal generators, including D , gauge rather than global charges. We call them *candidate physical classes under full residual conformal gauging*.

No statement in this paper establishes quantum BRST nilpotency, anomaly cancellation, interacting pseudo-unitarity, or a positive asymptotic particle space.

The local-to-residual bridge belongs to the general subject of local BRST cohomology with antifields, gauge fixing, and descent [12, 13]. Metsaev’s ordinary-derivative BRST–BV construction gives a concrete realization of conformal fields and their conformal algebra on fields and antifields [14]. These works do not supply the cylinder zero-mode transfer used here, but they delimit what must be proved before the conditional bridge can be promoted to a strict pure-Weyl theorem.

2 The free local pure-Weyl complex

2.1 Gauge, curvature, and equation maps

Let $h_{\mu\nu}$ be a metric perturbation and let (ξ^μ, σ) be an infinitesimal $\text{Diff} \times \text{Weyl}$ parameter. The full linear gauge map is

$$K(\xi, \sigma)_{\mu\nu} = \mathcal{L}_\xi \bar{g}_{\mu\nu} + 2\sigma \bar{g}_{\mu\nu} = 2\bar{\nabla}_{(\mu} \xi_{\nu)} + 2\sigma \bar{g}_{\mu\nu}. \quad (4)$$

After quotienting the trace, it becomes the conformal Killing operator

$$(K_0 \xi)_{\mu\nu} = 2\bar{\nabla}_{(\mu} \xi_{\nu)} - \frac{1}{2} \bar{g}_{\mu\nu} \bar{\nabla} \cdot \xi. \quad (5)$$

Let

$$C_1 h = \left. \frac{d}{d\varepsilon} C[\bar{g} + \varepsilon h] \right|_{\varepsilon=0} \quad (6)$$

be the linearized Weyl-curvature map, and let B_{lin} denote the action-normalized linearized Bach operator. Formal adjoints use the spacetime action pairing on a domain for which temporal and spatial boundary terms vanish.

The conformal cylinder is conformally flat. Naturality and conformal covariance therefore give the exact local identity

$$\boxed{C_1 K = 0} \quad (7)$$

for arbitrary local gauge parameters, not only for the fifteen residual conformal Killing parameters.

Proposition 2.1 (Action-normalized factorization). *Use the repository normalization*

$$S_{\text{red}}[g] = \int \sqrt{-g} \left(R_{\mu\nu} R^{\mu\nu} - \frac{1}{3} R^2 \right) = \frac{1}{2} \int \sqrt{-g} (C^2 - E_4). \quad (8)$$

On a conformally flat background, modulo the Euler boundary term,

$$\boxed{B_{\text{lin}} = C_1^\sharp C_1}. \quad (9)$$

Consequently,

$$B_{\text{lin}} K = 0, \quad K^\sharp B_{\text{lin}} = 0, \quad B_{\text{lin}}^\sharp = B_{\text{lin}}. \quad (10)$$

Proof. Since $C[\bar{g}] = 0$, the mixed second variation of (8) is

$$\delta_h \delta_k S_{\text{red}} = \langle C_1 h, C_1 k \rangle_{\mathcal{W}}. \quad (11)$$

The definition $\delta S_{\text{red}} = \langle h, \mathcal{E}(g) \rangle$ and $B_{\text{lin}} = D\mathcal{E}|_{\bar{g}}$ gives $\langle h, B_{\text{lin}} k \rangle = \langle C_1 h, C_1 k \rangle_{\mathcal{W}}$, proving (9). Equation (7) then gives the first detour identity, formal self-adjointness gives the second, and Hessian symmetry gives the third. $\square$

The Weyl-fibre pairing in Lorentz signature is indefinite. Thus $C_1^\sharp C_1$ is a formal-adjoint factorization, not a positivity factorization. In trace-free conformal geometry the corresponding local sequence is

$$\Gamma(T) \xrightarrow{K_0} \Gamma(S_0^2 T^*) \xrightarrow{B_{\text{lin}}} \Gamma(S_0^2 T^*) \xrightarrow{K_0^\sharp} \Gamma(T^*). \quad (12)$$

On Bach-flat backgrounds this is the conformal deformation detour complex of [1]. Formal self-adjointness holds in all signatures; ellipticity is a Riemannian statement and is not assumed on the Lorentzian cylinder.

2.2 What the local identities do not yet prove

The middle local cohomology relevant to the free field is

$$\frac{\ker B_{\text{lin}}}{\text{im } K}. \quad (13)$$

Equations (7)–(10) prove that this quotient is defined, but not that it is globally isomorphic to a prescribed normalizable cylinder module. In particular, the implication

$$\frac{\ker B_{\text{lin}}}{\text{im } K} \stackrel{?}{\cong} \mathcal{W}_+ \oplus \mathcal{W}_- \quad (14)$$

requires a Lorentzian domain analysis, an explicit curvature intertwiner, and removal of all local ghost and nonminimal doublets.

An exact finite-jet calculation gives strong evidence without promoting it to an all-level theorem. On homogeneous Euclidean polynomial jets of degree $n = 2, \dots, 6$, the matrices for K , C_1 , and B_1 obey

$$C_1 K = 0, \quad B_1 K = 0, \quad \ker C_1 = \text{im } K. \quad (15)$$

The exact ranks are shown in Table 1.

n	$\dim G_n$	$\dim h_n$	$\dim W_{n-2}$	$\dim B_{n-4}$	$\text{rank } K$	$\text{rank } C_1$	$\text{rank } B_1$	$\dim(\ker B_1 / \text{im } K)$
2	90	100	10	0	90	10	0	10
3	160	200	40	0	160	40	0	40
4	259	350	100	10	259	91	9	82
5	392	560	200	40	392	168	32	136
6	564	840	350	100	564	276	74	202

Table 1: Exact homogeneous-polynomial detour ranks. The calculation is a finite Euclidean jet certificate, not the missing all-level Lorentzian cylinder intertwiner.

The quotient dimensions (10, 40, 82, 136, 202) agree with the first five levels of the parity-complete $E/A/L$ module introduced below. The same calculation separates a 15-dimensional kernel of the low-degree gauge map, generated by four translations, six rotations, one dilation, and four special conformal transformations. These are exact finite results. The all-level version of (14) remains the principal local bridge stated in Section 8.

2.3 A prospective Lorentzian proof route

The missing implication is global and hyperbolic, not another formal detour identity. A proof must first choose its cylinder domain—for example, smooth finite-energy solutions with a specified frequency completion—because spacelike-compact, distributional, and unrestricted smooth solution spaces need not have the same quotient. The natural route is then:

1. gauge-fix the linear Bach complex so that its fourth-order wave operator factors into normally hyperbolic second-order factors, while isolating the conformal-Killing zero modes;
2. construct advanced/retarded homotopies on the nonzero-mode complex, using the framework of Green-hyperbolic complexes [3];
3. prove that C_1 is injective on the resulting metric quotient and surjective onto the chosen on-shell curvature module, with the Bianchi and Bach identities imposed;
4. separate the fifteen global reducibility parameters rather than burying them in local exactness, as required by the general treatment of global reducibility modules [15]; and
5. use the character (17), BGG structure [4], and the finite ranks above only as completeness checks.

Ordinary-derivative BRST–BV formulations of conformal fields provide a concrete field/antifield language for the gauge-fixed step [14]. None of these references by itself proves the Lorentzian cylinder quotient (14); the list records a proof programme, not an additional premise of the exact residual theorem.

3 The on-shell Weyl module

3.1 Character and cylinder towers

We now define the coefficient module used in the exact residual calculation. A positive-chirality Weyl curvature is a conformal primary with labels

$$(\Delta; j_L, j_R) = (2; 2, 0), \quad (16)$$

and the opposite chirality is its parity conjugate. The Bach equation is in $(4; 1, 1)$ and its differential identity is in $(5; \frac{1}{2}, \frac{1}{2})$. The resulting one-chirality character is

$$\chi_+(q) = \chi_{(2;2,0)} - \chi_{(4;1,1)} + \chi_{(5;\frac{1}{2},\frac{1}{2})} = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}. \quad (17)$$

This is the character of the on-shell Weyl-curvature module $\mathcal{W}_+$. Character equality is not, by itself, a proof of the local metric detour isomorphism (14).

On $\mathbb{R} \times S^3$, $\mathcal{W}_+$ decomposes into three multiplicity-free towers:

$$E_n^+ : \left(\frac{n+2}{2}, \frac{n-2}{2} \right), \quad \dim E_n = (n+3)(n-1), \quad n \geq 2, \quad (18)$$

$$A_n^+ : \left(\frac{n}{2}, \frac{n-2}{2} \right), \quad \dim A_n = (n+1)(n-1), \quad n \geq 3, \quad (19)$$

$$L_n^+ : \left(\frac{n}{2}, \frac{n-4}{2} \right), \quad \dim L_n = (n+1)(n-3), \quad n \geq 4. \quad (20)$$

The negative-chirality towers exchange j_L and j_R . Summing (18)–(20) reproduces (17). For both chiralities the first levels have dimensions

$$\dim \mathcal{H}_2 = 10, \quad \dim \mathcal{H}_3 = 40, \quad \dim \mathcal{H}_4 = 82, \quad \dim \mathcal{H}_5 = 136, \quad \dim \mathcal{H}_6 = 202. \quad (21)$$

The labels E, A, L are cylinder oscillator labels only. In particular, they should not be interpreted as three independently signable conformal representations. Proper conformal transformations mix the towers.

3.2 Invariant form and exact generator action

In normalized cylinder coordinates the unique standard-real, parity-compatible invariant form is, up to overall convention,

$$J_{\text{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \quad (22)$$

It is nondegenerate and indefinite. The proper conformal lowering and raising generators satisfy

$$(K_q^-)^\sharp = K_q^+, \quad X^\sharp = X \quad (X = D, SU(2)_L, SU(2)_R), \quad (23)$$

where $X^\sharp = J_{\text{conf}}^{-1} X^\dagger J_{\text{conf}}$.

Multiplicity one reduces each allowed lowering block to a scalar reduced coefficient. With source energy n , the six stable families are

$$\begin{aligned}
c_{EE}(n) &= \sqrt{\frac{2(n-1)(n+1)(n+3)}{n+2}}, & c_{AE}(n) &= \sqrt{\frac{8(n-1)}{(n-2)(n+2)}}, \\
c_{AA}(n) &= \sqrt{\frac{2(n-3)(n-1)(n+2)}{n-2}}, & c_{LE}(n) &= -\sqrt{\frac{2(n-3)}{n-2}}, \\
c_{LA}(n) &= \sqrt{\frac{8}{n-2}}, & c_{LL}(n) &= \sqrt{2(n-2)(n+1)}.
\end{aligned} \tag{24}$$

Together with Clebsch–Gordan coefficients and (23), these determine the all-level one-particle $\mathfrak{so}(4, 2)$ action. The exact reconstruction verifies the conformal commutators on every interior level of a finite energy buffer; the top shell is used only as a buffer and is never mistaken for a finite conformal representation.

The free Fock coefficient module is the bosonic second quantization

$$\mathcal{F}_{\mathcal{W}} = \text{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-), \tag{25}$$

with multiplicative lift of (22). The residual generators preserve oscillator particle number.

4 Hamiltonian moment-map normalization

The residual action is not introduced as an arbitrary matrix representation. In oscillator coordinates z , write the symplectic form as

$$\Omega = i d\bar{z} J \wedge dz. \tag{26}$$

If a conformal generator X acts linearly by $\delta_X z = K_X z$, its quadratic Hamiltonian moment map is

$$\mu_X(z, \bar{z}) = \bar{z} M_X z, \quad M_X = J K_X, \quad d\mu_X = \iota_{X_X} \Omega. \tag{27}$$

The J -adjoint identities ensure the correct reality of μ_X .

In the curved-cylinder wave normalization used for the independent Taub calculation, comparison with the quadratic action fixes the proper-conformal kernel to

$$M_{\text{Taub}} = -\frac{\sqrt{2}}{4\pi} J K^- \tag{28}$$

up to the explicitly declared orientation and harmonic phases. Two independent curvature components fix the common coefficient; it is not a one-entry fit. An exact two-chirality jet through source energy four verifies all fifteen moment-map components and the full conformal algebra on interior energies two and three.

The terms *moment map* and *Taub charge* refer to related but not identical constructions. Equation (27) is the Hamiltonian moment map of the residual linear action on oscillator phase space. A Taub charge is the pairing of a quadratic field-equation source with an adjoint zero mode; on a compact Cauchy surface it is a second-order integrability condition for extending a linearized perturbation to a nonlinear solution [16, 17]. The local zero-mode transfer and the chosen boundary problem are what identify the two. The finite calculation leading to (28) verifies

that identification for the tested proper-conformal kernels. It does not prove that every residual constraint in the full strict pure-Weyl BV problem is a Taub constraint, nor that declaring all residual charges gauge is forced.

This section supplies the Hamiltonian normalization of the residual action and its Taub interpretation. The cohomology theorem below depends only on the exact module action, not on interpreting a finite moment-map jet as a physical quotient.

5 Residual BRST and Cartan homotopy reduction

5.1 The absolute residual complex

Let

$$\mathfrak{g} = \mathfrak{so}(4, 2) \quad (29)$$

with compact grading

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \quad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4). \quad (30)$$

Here D and $\mathfrak{so}(4)$ lie in $\mathfrak{g}_0$, while proper conformal lowering and raising generators lie in $\mathfrak{g}_{-1}$ and $\mathfrak{g}_{+1}$. The dual ghosts carry the opposite compact degree.

For a coefficient module $V \subset \mathcal{F}_{\mathcal{W}}$, the absolute residual complex is

$$C^q(V) = V \otimes \Lambda^q \mathfrak{g}^*, \quad (31)$$

with Chevalley–Eilenberg differential

$$d = c^a \rho(G_a) - \frac{1}{2} f^a_{bc} c^b c^c \iota_a. \quad (32)$$

It raises ghost number by one and preserves the total compact degree

$$\delta = D_{\text{matter}} + D_{\text{ghost}}. \quad (33)$$

The standard centered polarization uses the product v of the four ghosts dual to the raising generators. It has

$$\text{gh}(v) = 4, \quad D_{\text{ghost}}(v) = -4. \quad (34)$$

This is the residual ghost convention used in the cylinder construction of [8]. Here it is an algebraic input. Its derivation from the strict pure-Weyl local BV zero-mode split is one of the hypotheses in Section 8; the broader Riegert sector of [8] is not included.

5.2 Full residual gauging is a physical assumption

Hamada’s background-independence programme identifies the transformations which preserve the cylinder gauge with residual diffeomorphisms and imposes the resulting conformal charges through BRST [7, 8]. This supplies a direct precedent for the complex (31). It does not prove that every quantization of strict pure-Weyl gravity must use the same quotient.

There are at least two distinct cylinder problems.

1. In the *fully gauged residual problem*, the cylinder is a gauge representative of a closed generally covariant system. All fifteen conformal-Killing transformations, including compact time translation, are generated by residual constraints. The ghost for D exists and the Cartan homotopy below is an allowed operation.

2. In the *fixed-background cylinder problem*, D is the Hamiltonian and $SO(4, 2)$ is a global symmetry acting on states or observables. Its eigenspaces are not quotiented. The D ghost and its contraction cannot be used to erase nonzero energy.

Because $[K^-, K^+]$ contains D , the second problem is not obtained by simply deleting one ghost while leaving the rest of (32) unchanged. It requires a different relative or boundary-charge complex. Theorem 6.1 answers only the first problem. A strict pure-Weyl derivation must state the temporal boundary conditions, the residual gauge group, and the treatment of its charges before calling the result a physical state space.

5.3 Cartan homotopy

Let ι_D be contraction with the generator D in the residual ghost exterior algebra. The usual Cartan identity is

$$d\iota_D + \iota_D d = \mathcal{L}_D. \quad (35)$$

Theorem 5.1 (Cartan homotopy reduction to compact degree zero). *Decompose the residual complex into total compact-degree eigenspaces,*

$$C = \bigoplus_{\delta} C_{\delta}, \quad \mathcal{L}_D|_{C_{\delta}} = \delta \text{ id}. \quad (36)$$

Every C_{δ} with $\delta \neq 0$ is contractible. Hence inclusion of the centered subcomplex is a quasi-isomorphism:

$$\boxed{H^{\bullet}(C, d) \cong H^{\bullet}(C_{\delta=0}, d)}. \quad (37)$$

Proof. On C_{δ} with $\delta \neq 0$, define

$$h_{\delta} = \frac{\iota_D}{\delta}. \quad (38)$$

Equation (35) gives

$$dh_{\delta} + h_{\delta}d = \text{id}. \quad (39)$$

Thus every closed vector of nonzero total degree has the explicit primitive $\delta^{-1}\iota_D\Psi$. If $\mathcal{L}_D$ has a nilpotent part, the same proof works wherever $\mathcal{L}_D$ is invertible by using $h = \iota_D\mathcal{L}_D^{-1}$. $\square$

The theorem depends on treating D as a residual gauge generator. If cylinder time translation is retained as a physical Hamiltonian or a boundary charge, contraction by its ghost is not an allowed quotient and (37) does not apply.

5.4 Finite centered inventory

At ghost number four, the minimal residual exterior algebra contains at most the four negative-degree ghosts, so

$$D_{\text{ghost}} \geq -4. \quad (40)$$

At $\delta = 0$ this implies

$$D_{\text{matter}} \leq 4. \quad (41)$$

Every nonvacuum Weyl quantum has weight at least two. Therefore only particle numbers $N = 0, 1, 2$ can contribute to the centered ghost degree selected by the Hamada polarization. All $N \geq 3$ sectors have matter weight at least six and are contractible before any matrix-rank computation.

6 Complete centered cohomology

6.1 The relative weight-four kernel

The complete matter Fock space at weight four is

$$\mathcal{F}_4 = \mathcal{H}_4 \oplus \text{Sym}^2 \mathcal{H}_2, \quad \dim \mathcal{F}_4 = 82 + 55 = 137. \quad (42)$$

The induced matter form has signature

$$\text{sig}(J_{\mathcal{F}_4}) = (97, 40). \quad (43)$$

The relative conformal-primary conditions are

$$(D - 4)|\Psi\rangle = 0, \quad R|\Psi\rangle = 0, \quad K_q^-|\Psi\rangle = 0. \quad (44)$$

Their exact kernel on the full space (42) is two-dimensional. In a spin-two magnetic basis its normalized chiral generators are

$$|W_{\pm}^2\rangle = \sqrt{\frac{2}{5}}|2, -2\rangle_{\pm} - \sqrt{\frac{2}{5}}|1, -1\rangle_{\pm} + \frac{1}{\sqrt{5}}|0, 0\rangle_{\pm}. \quad (45)$$

They obey

$$\langle W_r^2 | J_{\mathcal{F}} | W_s^2 \rangle = \delta_{rs}. \quad (46)$$

This relative computation is not by itself an absolute cohomology theorem; incoming exact states must also be excluded.

There is a simple representation-theoretic reason for the number two. The weight-two one-particle space is

$$E_2^+ \oplus E_2^- = (2, 0) \oplus (0, 2). \quad (47)$$

The symmetric square decomposes as

$$\text{Sym}^2(2, 0) \oplus [(2, 0) \otimes (0, 2)] \oplus \text{Sym}^2(0, 2). \quad (48)$$

There is one compact scalar in each same-chirality symmetric square and no scalar in the mixed product. Since $E_2^{\pm}$ are lowest-weight spaces, every K_q^- annihilates both factors, so these two scalar singlets are relative primaries. By contrast, the one-particle weight-four space contains

$$(3, 1) \oplus (2, 1) \oplus (2, 0) \quad (49)$$

and its parity conjugate, with no compact scalar. Thus representation theory predicts the two relative candidates before the matrix calculation. The absolute computation below is still necessary to exclude exactness and to prove the one-particle acyclicity. A complete Hochschild–Serre or Kostant-style derivation of the latter ranks would be a useful conceptual replacement for the large matrices [4].

6.2 Absolute kernel and image

The cutoff-complete absolute calculation supplies that missing test. For one chirality in the one-particle coefficient module, the centered complex around ghost number four has exact dimensions

$$C_0^3(290) \xrightarrow{d_3} C_0^4(1311) \xrightarrow{d_4} C_0^5(3657). \quad (50)$$

The entries lie in $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5})$. Good-prime reduction modulo 241, combined with exact nilpotency, fixes the characteristic-zero ranks to

$$\text{rank } d_3 = 260, \quad \text{rank } d_4 = 1051. \quad (51)$$

Since the ranks sum to 1311,

$$H_{\delta=0, N=1}^4 = 0. \quad (52)$$

Parity supplies the identical result for the opposite chirality.

The lowest two-particle block begins at matter weight four. Exterior saturation makes its beginning

$$0 \longrightarrow C_0^4(55) \xrightarrow{d_4} C_0^5(385) \xrightarrow{d_5} C_0^6(1155). \quad (53)$$

There is no incoming C_0^3 space: three residual ghosts have compact degree at least -3 and cannot center a two-particle coefficient of minimum weight four. Exact nilpotency gives $d_5 d_4 = 0$. The two vectors (45) lie in $\ker d_4$, while a good-prime minor gives $\text{rank } d_4 \geq 53$. The displayed independent kernel vectors give $\text{rank } d_4 \leq 53$. Therefore

$$H_{\delta=0, N=2}^4 = \text{span}\{[W_+^2], [W_-^2]\}. \quad (54)$$

For the vacuum coefficient module the complex is ordinary Lie-algebra cohomology. Since

$$H^\bullet(\mathfrak{so}(4, 2); \mathbb{C}) \cong \Lambda(u_3, u_5, u_7), \quad (55)$$

one has

$$H_{\delta=0, N=0}^4 = 0. \quad (56)$$

6.3 The normalized residual representative Gram matrix

The absolute ghost exterior algebra carries a complementary-degree pairing. The centered ghost representative v in (34) is normalized, using the standard residual insertion, so that

$$(v, v)_{\text{gh}} = 1. \quad (57)$$

This fixed insertion is not a nondegenerate inner product on each individual ghost-number block. It is sufficient here because the incoming two-particle space is empty and the two normalized representatives are fixed. Multiplying (57) by the matter restriction (46) gives their exact normalized residual representative Gram matrix. It is Hermitian and sesquilinear on the complex oscillator module; the parity-compatible real structure gives its real restriction. We reserve the phrase “transferred BV cohomological pairing” for the cyclic-transfer statement in Section 8.

Theorem 6.1 (Complete cohomology under full residual conformal gauging). *Let the coefficient module be the free bosonic Fock space $\mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-)$, let the residual algebra be $\mathfrak{so}(4, 2)$, treat D*

as a gauge generator, and use the centered ghost polarization (34). Then the complete minimal absolute residual cohomology at the selected ghost number four is

$$\boxed{H_{\text{res,min}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{rep}} = I_2.} \quad (58)$$

There is no one-particle class and no higher-matter-weight class in this minimal centered polarization.

Proof. The Cartan homotopy, Theorem 5.1, eliminates every nonzero total compact degree. The ghost-degree floor then reduces the centered inventory to $N = 0, 1, 2$. Equations (56), (52), and (54) exhaust those sectors. Equations (46) and (57) give the pairing. $\square$

Remark 6.2 (What is positive). The theorem does not turn the indefinite one-particle module (22) into a positive particle space. It removes the one-particle sector from this absolute residual cohomology. The positive I_2 belongs to two ghost-dressed scalar vertex classes.

7 Descent and the dynamical/topological split

7.1 Why weight four is selected

Let V_h be a scalar conformal primary of weight h , and let

$$\omega = \frac{1}{4!} \epsilon_{\mu\nu\rho\sigma} c^\mu c^\nu c^\rho c^\sigma \quad (59)$$

be the top lowering-ghost product. In the residual descent conventions,

$$sV_h = c^\mu \nabla_\mu V_h + \frac{h}{4} (\nabla_\mu c^\mu) V_h, \quad s\omega = -\omega \nabla_\mu c^\mu, \quad \omega c^\mu = 0. \quad (60)$$

It follows that both the integrated density and the unintegrated ghost-dressed operator have the same closure defect, proportional to $h/4 - 1$. Hence

$$\boxed{[\omega V_4] \longleftrightarrow \left[\int_M d^4x \sqrt{-g} V_4 \right]}. \quad (61)$$

The -4 compact degree of the ghost factor is therefore the conformal counterpart of the four-dimensional volume form, rather than an accidental normal-ordering shift. This weight-four state-operator mechanism and a Weyl-square example were already explicit in [8]; the new use here is the absolute exhaustion and pairing calculation of Section 6.

Three cohomological quotients appear and must not be identified without the transfer theorem:

$$H^4(\mathfrak{so}(4, 2); \mathcal{F}_{\mathcal{W}}), \quad H_{\text{loc}}^{0,4}(s \mid d), \quad \frac{\{\text{integrated local functionals}\}}{\{\text{allowed boundary functionals}\}}. \quad (62)$$

The first is the residual Chevalley–Eilenberg group computed exactly in this paper. The second is local BRST cohomology modulo spacetime descent [12]. The third depends on topology and boundary conditions. Equation (61) identifies representatives in the residual construction; its promotion to an isomorphism with the strict pure-Weyl local group is part of the conditional bridge.

7.2 Parity basis

Parity exchanges W_+^2 and W_-^2 . Define

$$e = \frac{W_+^2 + W_-^2}{\sqrt{2}}, \quad o = \frac{W_+^2 - W_-^2}{\sqrt{2}}. \quad (63)$$

Up to the orientation-dependent factor of i in Lorentzian self-duality conventions, descent identifies

$$e \sim C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}, \quad o \sim C_{\mu\nu\rho\sigma} \tilde{C}^{\mu\nu\rho\sigma}. \quad (64)$$

Because (63) is orthogonal, its Gram matrix remains I_2 .

7.3 Euler–Lagrange quotient

Let

$$\mathcal{E} : H_{\text{res,min}}^4 \longrightarrow \{\text{local equation and gauge deformations}\} \quad (65)$$

be the Euler–Lagrange map obtained by varying the integrated density. In the conventions of Section 2,

$$\mathcal{E}(e) = \lambda_B B_{\mu\nu}, \quad \lambda_B \neq 0, \quad \mathcal{E}(o) = 0. \quad (66)$$

The first identity is the Weyl-square variation. Under locality, smoothness, Lorentz invariance, and a four-derivative bound, it is the unique nontrivial self-interaction of a single linear conformal graviton modulo equivalence [11].

The second identity is Chern–Weil transgression. In a local frame,

$$P_4 = \text{Tr}(R \wedge R) = dQ_3(\Gamma), \quad (67)$$

where

$$Q_3(\Gamma) = \text{Tr} \left(\Gamma \wedge d\Gamma + \frac{2}{3} \Gamma \wedge \Gamma \wedge \Gamma \right). \quad (68)$$

Its variation is

$$\delta P_4 = 2d \text{Tr}(\delta\Gamma \wedge R), \quad (69)$$

using the Bianchi identity. On a finite cylinder, in a chosen global frame,

$$\int_{[t_1, t_2] \times S^3} P_4 = \text{CS}[\Gamma(t_2)] - \text{CS}[\Gamma(t_1)]. \quad (70)$$

Thus fixed-endpoint or compactly supported variations have no bulk Euler–Lagrange derivative.

Equations (66)–(70) give the exact sequence

$$0 \longrightarrow \text{span}\{o\} \longrightarrow H_{\text{res,min}}^4 \xrightarrow{\mathcal{E}} \text{span}\{B_{\mu\nu}\} \longrightarrow 0. \quad (71)$$

Quotienting variationally trivial local bulk functionals yields

$$H_{\text{dyn}}^4 = H_{\text{res,min}}^4 / \ker \mathcal{E} \cong \text{span}\{[C^2]\}, \quad \boxed{G_{\text{dyn,rep}} = I_1}. \quad (72)$$

Equivalently,

$$\boxed{I_2 = I_1^{\text{dynamical}} \oplus I_1^{\text{topological}}}. \quad (73)$$

The odd class is only locally and variationally trivial. A gravitational theta parameter can affect boundaries, horizons, contact terms, large frame transformations, and nontrivial topology [19]. We do not quotient those global data without specifying the boundary problem.

8 The local-to-residual transfer

Theorem 6.1 begins with the locally reduced Weyl module. This section states exactly what is needed to identify it with the complete free pure-Weyl BV answer.

8.1 Equivariant Cartan transfer

Let (C, q) be the gauge-fixed local free BV complex after the fifteen conformal-Killing zero modes have been separated, and let $H = H(q)$. Suppose there is a strong deformation retract

$$(H, 0) \begin{smallmatrix} \xrightarrow{j} \\ \xleftarrow{p} \end{smallmatrix} (C, q), \quad jp = \text{id} - qs - sq, \quad pj = \text{id}_H. \quad (74)$$

Let Δ contain the residual zero-mode differential and mixed terms, so $Q = q + \Delta$. With the sign convention in (74), homological perturbation gives

$$I = (\text{id} + s\Delta)^{-1}j, \quad P = p(\text{id} + \Delta s)^{-1}, \quad Q_H = p\Delta(\text{id} + s\Delta)^{-1}j. \quad (75)$$

If

$$[D, q] = [D, \Delta] = [D, s] = 0, \quad pD = D_H p, \quad Dj = jD_H, \quad (76)$$

then the transferred operators

$$\iota_D^H = P\iota_D I, \quad \mathcal{L}_D^H = P\mathcal{L}_D I \quad (77)$$

obey

$$[Q_H, \iota_D^H]_+ = \mathcal{L}_D^H. \quad (78)$$

Thus every transferred subspace on which $\mathcal{L}_D^H$ is invertible is again contractible.

8.2 Cyclic transfer and the pairing

Let $\sharp$ be the graded adjoint for the full BV/Krein pairing. Suppose the unperturbed retract is cyclic,

$$j^\sharp = p, \quad (s\Delta)^\sharp = \Delta s. \quad (79)$$

Then taking the adjoint of (75) gives

$$I^\sharp = P. \quad (80)$$

Since $PI = \text{id}_H$,

$$\boxed{I^\sharp I = \text{id}_H.} \quad (81)$$

The dressed inclusion is therefore an exact isometry: contractible local dressing changes representatives without changing the reduced Gram matrix.

In finite compact-degree blocks, a cyclic retract exists algebraically when $q^\sharp = -q$ and the induced cohomology pairing is nondegenerate. One chooses a nondegenerate representative space $\mathcal{H} \subset \ker q$, an isotropic complement to $\text{im } q$, and defines s as the inverse of q on that complement. The remaining field-theory obligation is to make this choice simultaneously compatible with compact degree, the cylinder domain, and the zero-mode split.

8.3 The filtered local-to-residual complex

The possible failure modes can be displayed without compressing them into the phrase “no other row.” After the conformal-Killing zero modes have been split from the local ghosts, introduce the residual-ghost filtration on the bicomplex

$$C^{p,q} = \Lambda^p \mathfrak{g}^* \widehat{\otimes} C_{\text{local}}^q, \quad F^p C = \bigoplus_{r \geq p} C^{r,*}. \quad (82)$$

Here p is residual ghost number and q is local BV degree. A transferred degree-one differential has the filtered expansion

$$Q_H = q_{\text{local}} + d_{\text{res}} + \sum_{k \geq 2} Q_{[k]}, \quad (83)$$

with bidegrees

$$|q_{\text{local}}| = (0, 1), \quad |d_{\text{res}}| = (1, 0), \quad |Q_{[k]}| = (k, 1 - k). \quad (84)$$

The terms $Q_{[k]}$ encode higher transferred ghost operations; assuming that they vanish is a substantive strictness assertion, not automatic homological bookkeeping.

The resulting spectral sequence begins

$$E_1^{p,q} = \Lambda^p \mathfrak{g}^* \otimes H^q(q_{\text{local}}), \quad E_2^{p,q} = H^p(\mathfrak{g}; H^q(q_{\text{local}})), \quad (85)$$

and

$$d_r : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}. \quad (86)$$

For the desired term $E_r^{4,0}$, a higher differential can enter from $(4-r, r-1)$ for $r = 2, 3, 4$, and can leave toward $(4+r, 1-r)$ for $2 \leq r \leq 11$. Thus local classes in degrees $q = 1, 2, 3$ and antifield or negative-degree classes down to $q = -10$ are potential modifiers. They must be inventoried or excluded; it is not enough to compute only the local degree-zero row. Since $0 \leq p \leq 15$, this filtration is bounded. On each finite compact-energy block the cochain spaces used here are finite, so the spectral sequence converges. An infinite completed state space additionally requires a complete, separated filtration. Even after collapse, a cyclic splitting is needed to control extension data and the Hermitian form.

This is the standard local-BRST setting in which antifield and gauge-fixed cohomologies must be compared carefully [12, 13].

Proposition 8.1 (Formal filtered-transfer criterion). *Suppose the zero-mode split produces (82); the filtration is convergent; $H^q(q_{\text{local}})$ vanishes in every row that can enter or leave $(4, 0)$; and the transferred differential on $H^0(q_{\text{local}})$ is the strict residual Chevalley–Eilenberg differential, with all relevant $Q_{[k]}$ absent. Then the full cohomology in the selected filtered degree is isomorphic, as a vector space, to the residual group*

$$H^4(\mathfrak{g}; H^0(q_{\text{local}})). \quad (87)$$

If in addition the retract is compact-degree equivariant and cyclic, the Cartan homotopy and the cohomological Hermitian pairing transfer, and the dressed inclusion is the isometry (81).

Proof. Under the stated row and strictness assumptions no differential (86) enters or leaves $(4, 0)$, so the bounded spectral sequence collapses there to its E_2 value. The equivariant and cyclic statements are (78) and (81). The cyclic splitting fixes the extension of the pairing from the associated graded object. $\square$

8.4 Pure-Weyl field-theoretic hypotheses

Field-Theoretic Hypothesis 8.2 (Strict pure-Weyl zero-mode transfer). The gauge-fixed free strict-pure-Weyl BV complex on the stated cylinder domain has the following properties.

1. Its local $\text{Diff} \times \text{Weyl}$ ghosts split exactly into the fifteen conformal-Killing zero modes and a nonzero-mode complement, without double counting either sector.
2. After nonminimal doublets are contracted,

$$H^0(q_{\text{local}}) \cong \mathcal{F}(\mathcal{W}_+ \oplus \mathcal{W}_-), \quad (88)$$

which includes the all-level Lorentzian cylinder isomorphism (14).

3. The local ghost and antifield rows which could enter or leave $E_r^{4,0}$ in (86) vanish, and the filtration is complete and convergent on the chosen function space.
4. A compact-degree-equivariant cyclic strong deformation retract exists and satisfies (76) and (79).
5. The transferred differential is the strict residual $\mathfrak{so}(4,2)$ Chevalley–Eilenberg differential (32); the higher operations $Q_{[k]}$ vanish or are removed by a filtered cyclic equivalence in the centered window.
6. The zero-mode measure induces the Hamada ghost polarization with the degree and normalization (34) and (57), and the parity-compatible real structure transfers.

These are field-theoretic hypotheses, not consequences of the exact residual matrices. In particular, Items 3 and 5 contain the two mechanisms by which the residual answer could fail to equal the full BV answer.

Corollary 8.3 (Conditional strict-pure-Weyl result). *If Field-Theoretic Hypothesis 8.2 holds, the complete free classical local-plus-residual BV cohomology in the selected degree is (58). Its transferred cohomological Hermitian form is I_2 , and its local Euler–Lagrange quotient is (72) with form I_1 .*

Proof. Items 1–3 identify the filtered complex and coefficient module required by Proposition 8.1. Items 4–5 transfer the Cartan homotopy, strict residual differential, and pairing. Item 6 identifies the selected degree and normalization. The exact residual theorem then gives (58), while (81) promotes its representative Gram matrix to the transferred cohomological form. $\square$

The formal detour theorem, factorization (9), fifteen-dimensional zero-mode count, and finite jet table are exact inputs. Separate finite-dimensional fixtures verify (78) and (81) with a genuinely dressed inclusion. They do not instantiate the full pure-Weyl field complex. The unproved core is the Lorentzian local-kernel theorem and the row/strictness assertions made explicit above.

9 Scope, interpretation, and open questions

Residual gauge choice. The result is specific to quotienting the full residual $SO(4,2)$ action on the compact cylinder. If some generators are retained as global or asymptotic charges, the physical complex is different. In particular, Cartan contraction cannot be used to erase nonzero D energy when D is the physical Hamiltonian.

No positive graviton Fock claim. The unreduced module remains indefinite and the one-particle residual cohomology is zero. The positive classes are weight-four scalar vertices. Calling I_2 a positive graviton Hilbert space would therefore misidentify both the cohomological degree and the state-operator map.

Classical and free. The result concerns the free classical complex and its finite residual reduction. It does not show that the interaction preserves the reduced space or pairing. Nor does it address the quantum master equation. A local Diff $\times$ Weyl anomaly could obstruct quantum nilpotency even if Corollary 8.3 is fully established.

Vertex versus dynamics. The two residual classes do not define two independent local gravitational dynamics. The parity-even class is the Weyl action direction; the parity-odd class is a theta direction, locally related by a canonical boundary transformation. The positive dynamical quotient is therefore one-dimensional.

The remaining classical theorem. The highest-priority mathematical task is the all-level Lorentzian cylinder identification

$$\ker B_{\text{lin}} / \text{im } K \cong \mathcal{W}_+ \oplus \mathcal{W}_- \quad (89)$$

inside the complete local BV complex, followed by a compact-degree equivariant cyclic zero-mode transfer. The exact ranks in Table 1 are a finite certificate and a regression rail, not a substitute for that proof.

10 Conclusion

The conventional free oscillator description of pure conformal gravity on $\mathbb{R} \times S^3$ is indefinite. Residual conformal reduction changes the question rather than re-signing those oscillators. The absolute residual complex carries a Cartan contraction which eliminates every noncentered compact degree. At the ghost number four selected by the centered polarization this turns the apparent infinite problem into the three sectors $N = 0, 1, 2$. The first two are acyclic; the last has exactly two classes and positive Gram matrix:

$$H_{\text{res,min}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{rep}} = I_2.$$

Descent identifies them as the Weyl-square and Pontryagin vertex directions, and the quotient by variationally trivial bulk functionals leaves one positive dynamical direction.

The result is both stronger and narrower than a sampled positivity calculation. It is exhaustive for the stated minimal residual complex, but it concerns vertices rather than propagating particles. Its promotion to the full free pure-Weyl BV theory is controlled by the explicit hypotheses of Corollary 8.3. Quantum consistency and interaction descent are separate questions.

A Conventions

The cylinder radius is one and the Lorentzian signature is $(-, +, +, +)$. Compact energy is the eigenvalue of $D = i\partial_t$ in the oscillator convention. The compact algebra is $SO(4) \cong SU(2)_L \times$

$SU(2)_R$. A label (j_L, j_R) has dimension $(2j_L + 1)(2j_R + 1)$. Positive chirality means the lowest Weyl primary $(2; 2, 0)$; parity exchanges j_L and j_R .

The normalized module convention is (22). The reduced curvature action (8) differs by an overall factor and sign from the common $-\int C^2$ oscillator convention. All moment-map comparisons derive the symplectic form and Taub kernel from the same action, so the overall action factor cancels. Lorentzian self-duality may insert a factor of i in the odd parity combination; no conclusion depends on that convention.

The residual generator degrees are $(0^7, -1^4, +1^4)$ and dual ghost degrees are their negatives. Ghost number and compact degree are independent. The selected absolute residual degree in this paper is ghost number four, fixed by (34); it is not ordinary ghost number zero. We call it a candidate physical degree only under the full residual gauging specified in Section 5.2.

B Exact rank certification

The one-particle matrices contain radicals. They are represented over

$$K = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5}). \quad (90)$$

For a lower bound on rank, the calculation maps this field to $\mathbb{F}_{241}$ by

$$i \mapsto 64, \quad \sqrt{2} \mapsto 22, \quad \sqrt{3} \mapsto 56, \quad \sqrt{5} \mapsto 103. \quad (91)$$

Each image obeys its defining square relation. A nonzero minor after this good-prime reduction is a nonzero characteristic-zero minor. Exact nilpotency provides the complementary upper bound

$$\text{rank } d_{q-1} + \text{rank } d_q \leq \dim C^q. \quad (92)$$

For the one-particle complex the modular lower bounds saturate this inequality. For the two-particle complex the two explicit independent kernel vectors provide the upper bound which the modular rank saturates. Thus the ranks quoted in Section 6 are exact characteristic-zero ranks, not probabilistic modular estimates.

C Claim-status and verification index

The scientific input was frozen at repository commit

`e928f257c25099eb534eb34109dfc1dc6a3127a1`.

The companion snapshot records the environment, hashes, expected-failure guards, and command lines. The complete paper verification is run with

`python3 symbolic/verify_conformal_paper_free.py`

Claim	Status	Evidence / boundary
Formal pure-Weyl detour identities	Exact	Branson–Gover theorem plus action-normalized factorization and finite scalar-block audits.
Euclidean polynomial detour ranks, degrees 2–6	Exact finite certificate	Rational matrices verify kernels, ranks, and the 15 residual zero modes; no all-level or Lorentzian identification is inferred.
On-shell Weyl module and stable generator coefficients	Exact module construction	Character, tower inventory, invariant form, and cutoff-stable conformal brackets.
Residual Cartan homotopy reduction	Theorem	Explicit homotopy ι_D/δ on every nonzero total degree; this is a quotient only in the full residual-gauging problem.
Centered vacuum and one-particle sectors	Exact	Semisimple Lie-algebra cohomology for the vacuum; cutoff-complete absolute rank certificate for one particle.
Centered two-particle sector and $G_{\text{rep}} = I_2$	Exact residual theorem	Independent relative and absolute calculations; empty incoming absolute space and exact rank 53.
Vertex descent and dynamical/topological split	Exact locally	Weight-four descent, Chern–Weil transgression, and Euler–Lagrange quotient. Global theta effects are retained.
Full local-plus-residual pure-Weyl BV result	Conditional	Requires Field-Theoretic Hypothesis 8.2; the all-level Lorentzian local-kernel theorem, row inventory, and strict transfer remain open.
Interpretation as a physical state space	Choice-dependent	Requires a boundary and gauge principle that treats all fifteen generators, including D , as gauge rather than global charges.
Quantum anomaly freedom and interacting probability	Not claimed	Outside the scope of this paper.

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